

WASP-2b

R-1448

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-2_220828 · created 2026-08-01T09:16:57+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459819.742 +/- 0.0072 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.16 +/- 0.045
Transit depth (Rp/Rs)^2	2.56 +/- 1.44 [%]
Orbital Inclination (inc)	85.34 +/- 2.87
Airmass coefficient 1 (a1)	0.141 +/- 0.057
Airmass coefficient 2 (a2)	1.74 +/- 0.9
Scatter in the residuals of the lightcurve fit is	43.17 %
Best Comparison Star	#3 - [126, 309]
Optimal Aperture	1.11
Optimal Annulus	6.5
Transit Duration (day)	0.072 +/- 0.031

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.16 ± 0.045 vs 0.1326 (deviation 0.61σ)
Duration fit vs published	0.072 d vs 0.0737 d (deviation 0.06σ)
Mid-time O–C	13.7 min
Depth SNR	3.6
Residual scatter	43.17 %

Light curve

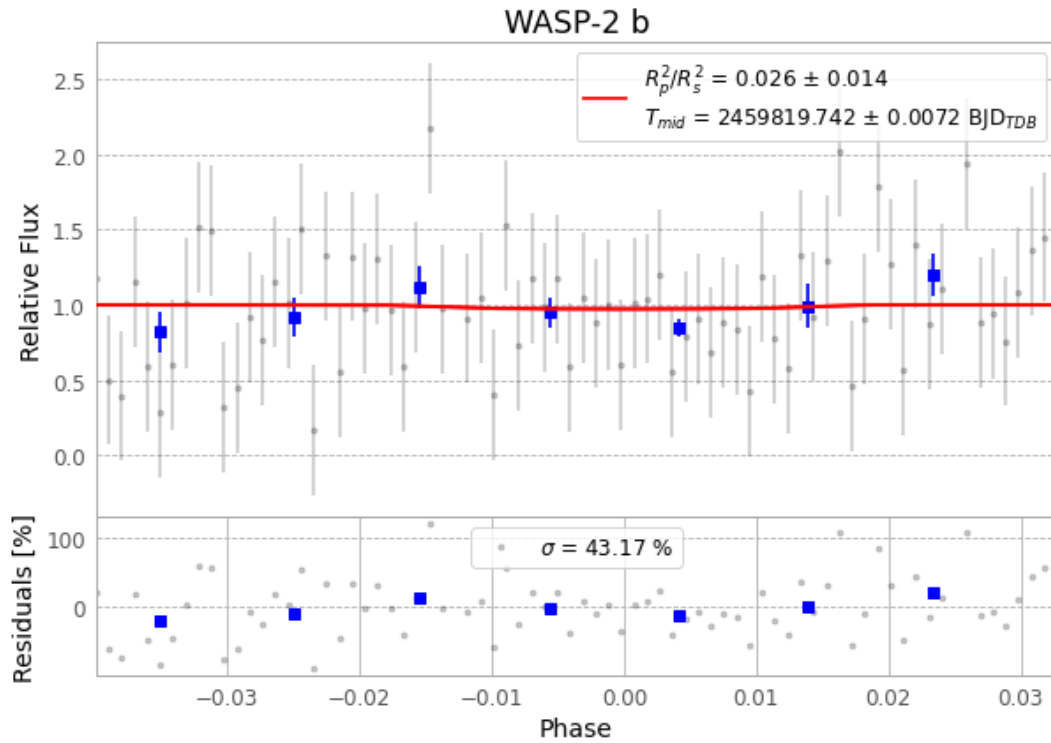


Figure 1 — FinalLightCurve_WASP-2 b_2022-08-27.png (SHA-256 0ffac0f1e1a9e9a9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [267, 54], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 12f30b4fe7e52e9b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

76 FITS frames (50 MB), WASP-2220828033613.FITS → WASP-2220828072113.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-220828.log (0316a0523cb1...), FinalLightCurve_WASP-2 b_2022-08-27.png (0ffac0f1e1a9...), FinalLightCurve_WASP-2 b_2022-08-27.csv (da027c04f37a...)

Compute resources

Wall time 205.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 09:16Z.